SKINGUARD AI

Explainable AI Based Dermoscopic Skin Analyzer Report

AI-assisted skin lesion screening and analysis report

1. Scan Information

Parameter	Result
Scan ID	SG-20260927134222
Date & Time	27-09-2026 13:42
Predicted Disease	Melanocytic Nevus
Confidence	17.98%
Severity	Low
Severity Score	17.98%

2. Uploaded Dermoscopic Image



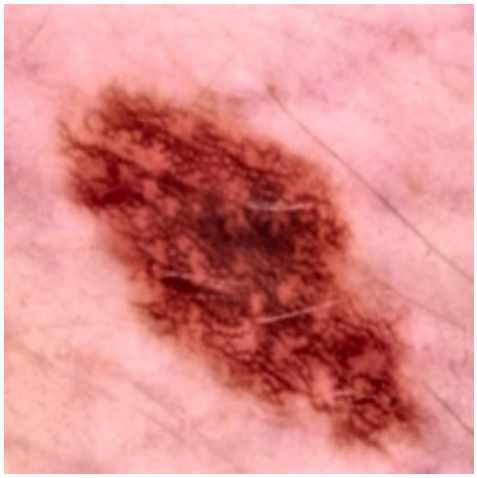
Original dermoscopic image uploaded for AI-based skin analysis.

3. Image Quality Check

Parameter	Result
Quality Score	95.00%
Blur	Not Available
Brightness	Not Available
Noise	Not Available
Overall Status	Excellent Image

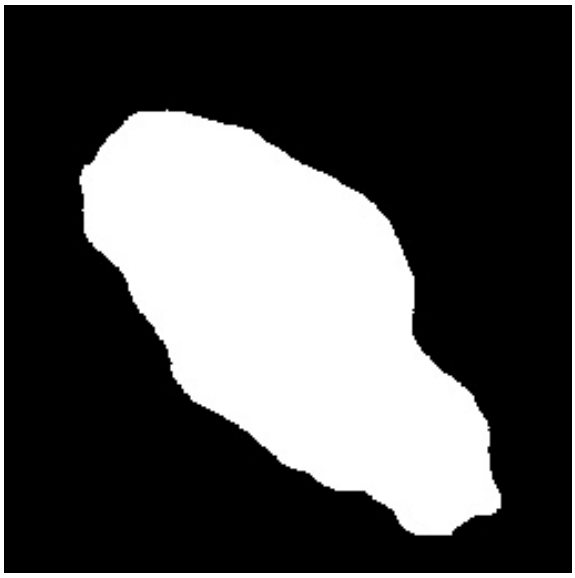
4. Image Pre-processing

Parameter	Result
Status	Completed
Resize	224 × 224 pixels
Enhancement	Applied
Normalization	Applied
Pipeline	Original → Resized → Enhanced → Normalized



5. Lesion Segmentation

Parameter	Result
Segmentation Model	U-Net
Status	Segmentation Completed Successfully
Lesion Area	Detected



U-Net based lesion segmentation output.

6. Disease Classification

Parameter	Result
Predicted Class	Melanocytic Nevus
Confidence Score	17.98%
Classification Model	EfficientNet-B3
Number of Classes	7

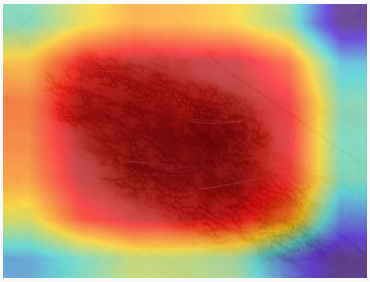
7. Seven-Class Probability Distribution

Disease Class	Probability
Actinic Keratosis	14.41%
Basal Cell Carcinoma	14.00%
Benign Keratosis	14.83%
Dermatofibroma	11.19%
Melanocytic Nevus	17.98%
Melanoma	14.96%
Vascular Lesion	12.62%

Probability distribution generated by the EfficientNet-B3 classification model.

8. Explainable AI - Grad-CAM

Grad-CAM: Grad-CAM highlights the image regions that contribute to the model prediction.

		
Original Image	Grad-CAM Heatmap	Important Region

9. Severity Prediction

Parameter	Result
Severity Level	Low
Severity Score	17.98%
Algorithm	Rule-Based Severity Prediction

Personalized Clinical Recommendation

Low severity has been detected. Continue regular skin monitoring and consult a dermatologist if any unusual changes are observed.

10. AI System Summary

System Component	Technology / Method
Image Pre-processing	OpenCV
Lesion Segmentation	U-Net
Disease Classification	EfficientNet-B3
Explainable AI	Grad-CAM
Severity Prediction	Rule-Based Algorithm
Application	Flask

MEDICAL DISCLAIMER

This AI-generated report is intended for screening and clinical decision support purposes only. It should not be considered as a substitute for professional medical diagnosis. Please consult a qualified dermatologist for final evaluation.

Report generated by SkinGuard AI on 27-09-2026 13:42